

Safety and Reliability Training

Session 1

Introduction to Systems Engineering



[Safety and Reliability Training](#)

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ANZEN

ANZEN PROPRIETARY INFORMATION

 **esa**
business incubation centre

 **fundación para el
conocimiento
madrid**
Comunidad de Madrid



Objectives of this training

- ✓ To understand the role and purpose of systems engineering
- ✓ Get to know the main phases of the development cycle of a system
- ✓ Get to know the main supporting documents and deliverables along the development cycle
- ✓ To understand the link between system engineering and safety engineering



Outline

- ❑ Introduction
- ❑ Fundamentals of systems engineering
 - What is a system?
 - System decomposition
 - System lifecycle
- ❑ The V-cycle applied to aerospace systems
 - Main phases of the development cycle of a system
 - Main supporting documents and deliverables of the development cycle
- ❑ Introduction to configuration management
- ❑ The V-cycle applied to system-safety engineering process
- ❑ Modern organizations: *waterfall methodology vs. agile methodology*

Introduction

Systems engineering – a key discipline to master complexity



Fuselage Group



Controls Group



Hydraulics Group



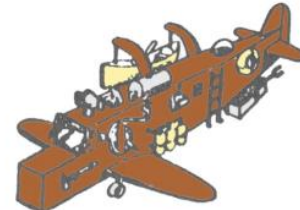
RM&S Group



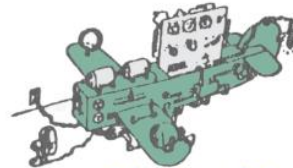
Wing Group



Empennage Group



Equipment Group



Electrical Group



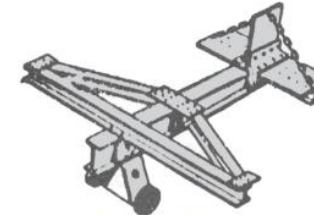
Powerplant Group



Armament Group



Aerodynamics Group



Stress Group



Weight Group



Loft Group



Production Engineering Group

Introduction

Systems engineers – the architects of complex products

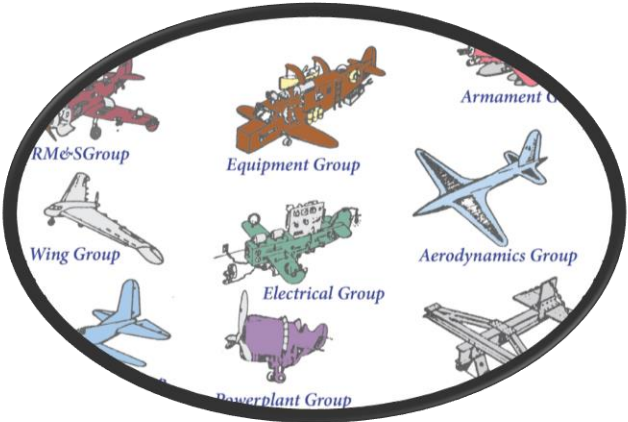
50 PASSENGER AIRCRAFT	
ATR 42	\$
Range: 1,555 km / 966 miles	
Seats: 50	Baggage capacity: 1,200 kg / 2,646 lbs
EMBRAER 145	\$\$
Range: 2,870 km / 1,783 miles	
Seats: 50	Baggage capacity: 1,000 kg / 2,205 lbs
BOEING 737-500 VIP	\$\$\$
Range: 5,970 km / 3,709 miles	
Seats: 56	Baggage capacity: 3,000 kg / 6,614 lbs
BOEING 757-200 VIP	\$\$\$\$
Range: 7,222 km / 4,487 miles	
Seats: 62-182	Baggage capacity: 5,000-6,000 kg / 11,023-13,228 lbs

Commercial

User needs



Design iterations



Technical disciplines

Systems Engineers



Introduction



International Council
on Systems Engineering

Systems engineering definition according to INCOSE

- Systems Engineering is a transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods.
- We use the terms “engineering” and “engineered” in their widest sense: “the action of working artfully to bring something about”. “Engineered systems” may be composed of any or all of people, products, services, information, processes, and natural elements.

Fundamentals of systems engineering

What is a system?

*System ?
Airplane*



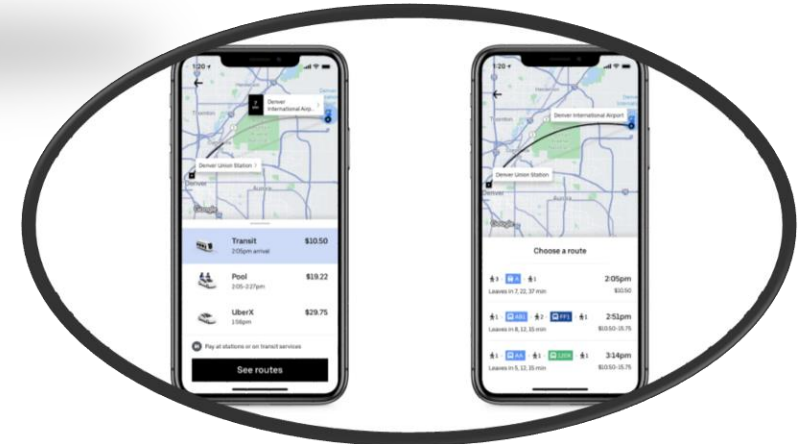
*System ?
Wind turbines*



*System ?
Car*



*System ?
Mobile app*



Fundamentals of systems engineering

What is a system?

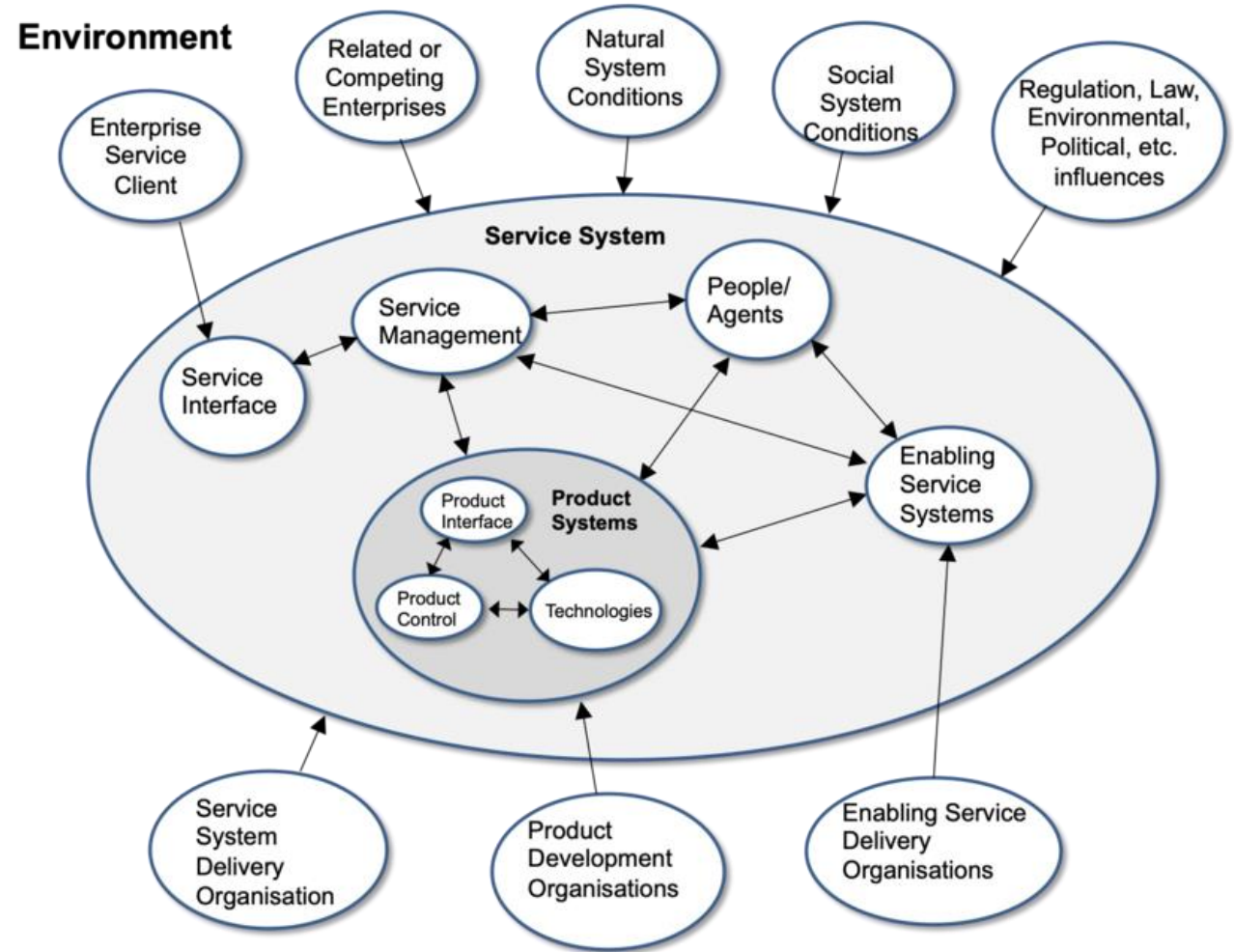


- *System* could be a very vague concept: an airplane is a system, the landing gear is a system, the mechanism deploying the landing gear is also a system...
- In the engineering context we should understand system as ***system-of-interest***

Fundamentals of systems engineering

What is a system? INCOSE definition

- A **system** is an arrangement of parts or elements that together exhibit behavior or meaning that the individual constituents do not.
- An **engineered system** is a system designed or adapted to **interact** with an anticipated operational **environment** to **achieve** one or more intended **purposes** while **complying** with applicable **constraints**.

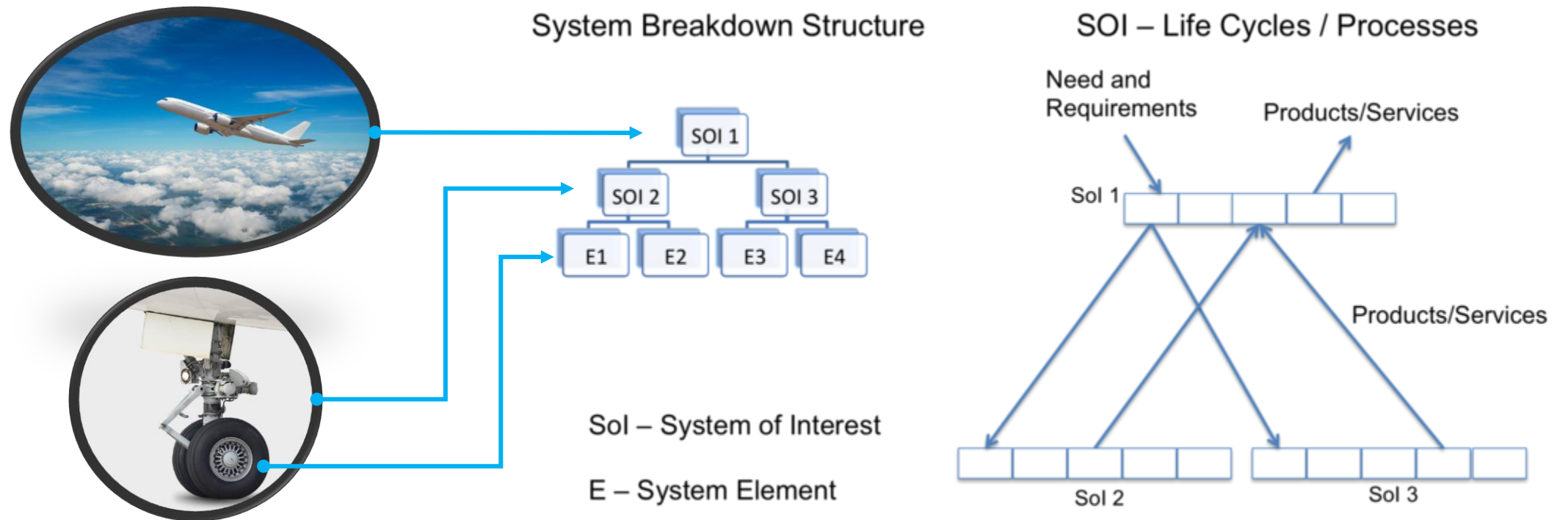


Source: SEBoK - Introduction to Systems Engineering Fundamentals

Fundamentals of systems engineering

System decomposition – System breakdown structure

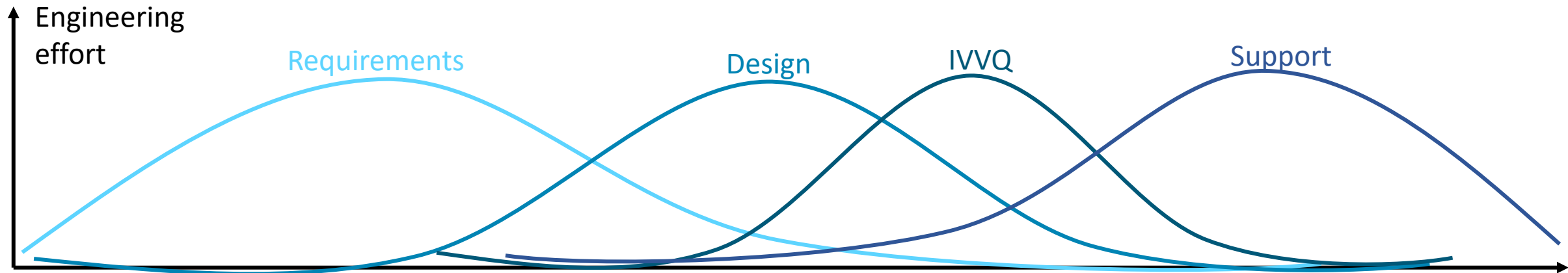
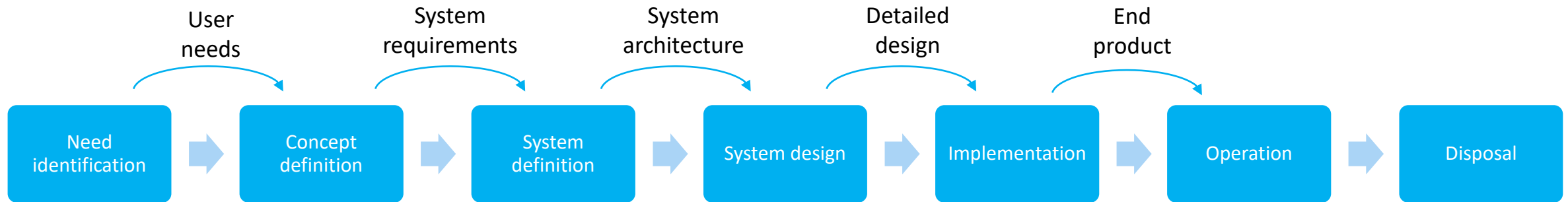
- Systematic decomposition of systems into smaller systems (element/parts/components)



[Source: SEBoK - Systems Lifecycle Approaches](#)

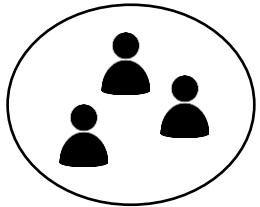
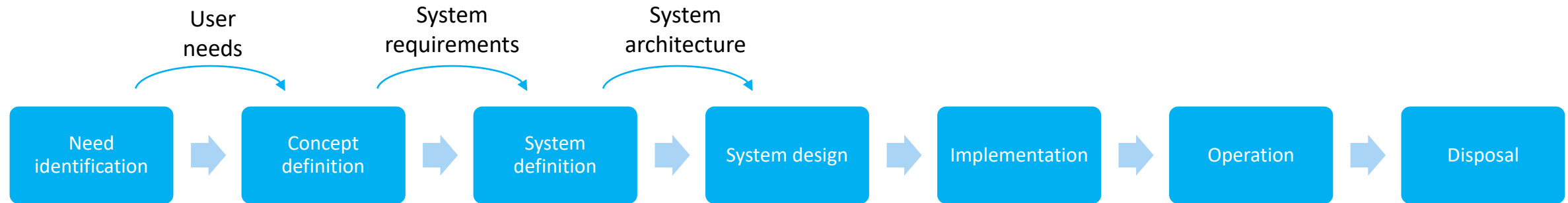
Fundamentals of systems engineering

System lifecycle



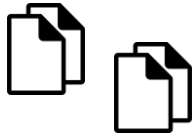
Fundamentals of systems engineering

System lifecycle – Stakeholder needs



Stakeholders – any individual/organization participating to the system lifecycle

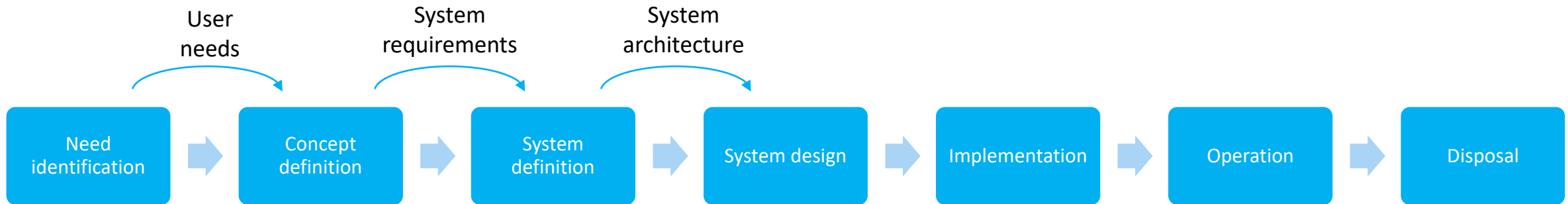
- Commercial
- Certification agencies
- Systems engineers
- Technical specialists
- Design engineers
- Suppliers
- Quality
- Sales
- End user
- Operation support
- Maintenance
- ...



Stakeholder needs – desired capabilities of the system – translated into **requirements**

Fundamentals of systems engineering

System lifecycle – Requirements

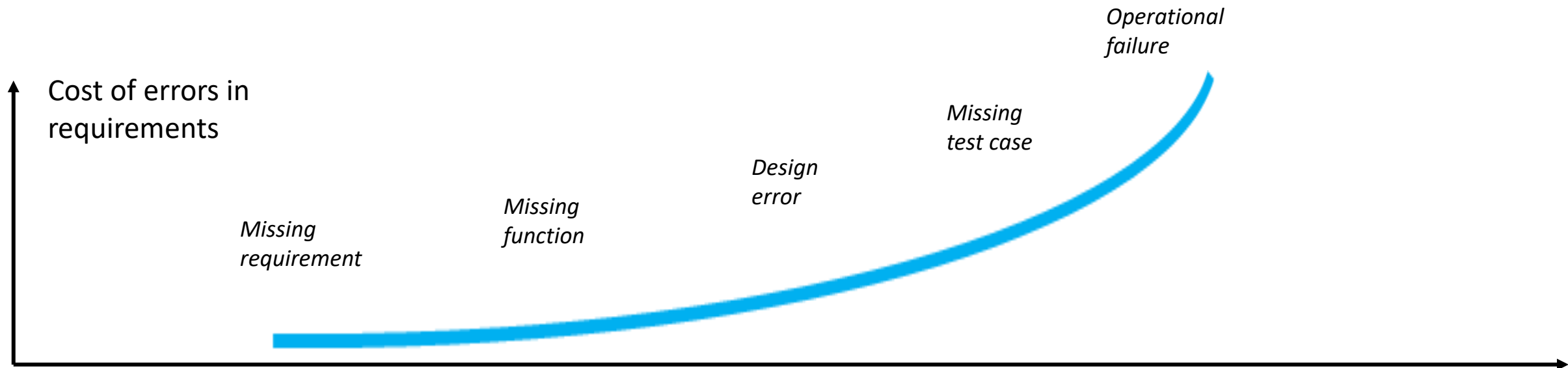
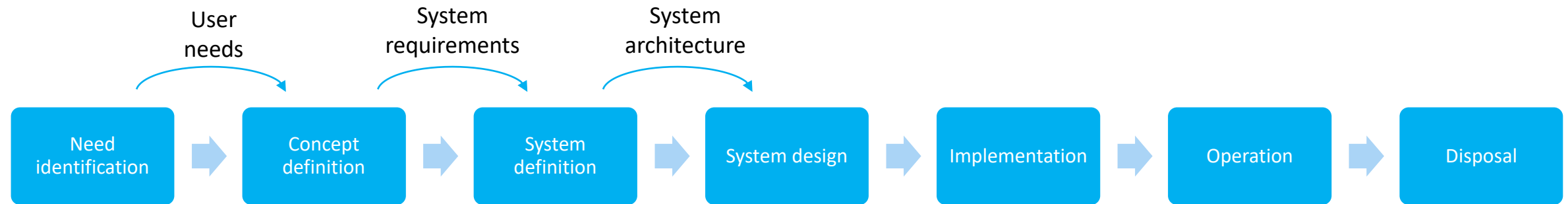


Requirements

- *Statement that identifies a product or process operational, functional, or design characteristic or constraint, which is **unambiguous, testable or measurable**, and **necessary** for product or process acceptability. (ISO/IEC 2007)*
 - Functional requirements
 - Performance reqs
 - Usability reqs
 - Interface reqs
 - Operational reqs
 - Modes reqs
 - Adaptability reqs
 - Physical constraints
 - Design constraints
 - Environmental conditions
 - Logistics
 - Regulations
 - Cost and schedule constraints
- *Requirement verification - the 3C's: correct, consistent, complete*

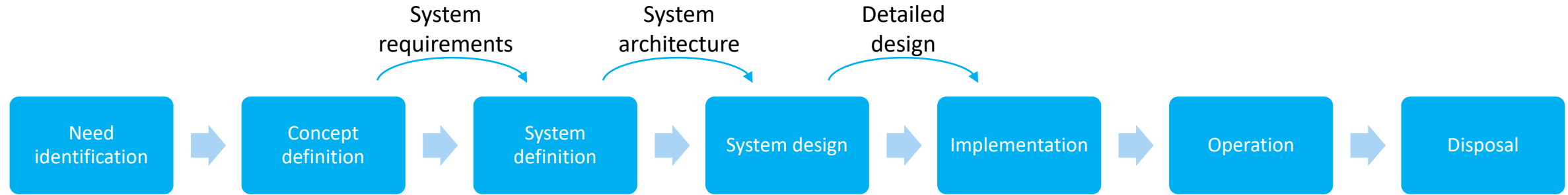
Fundamentals of systems engineering

System lifecycle – Requirements



Fundamentals of systems engineering

System lifecycle – system design



Step 1 – Operational/functional Analysis

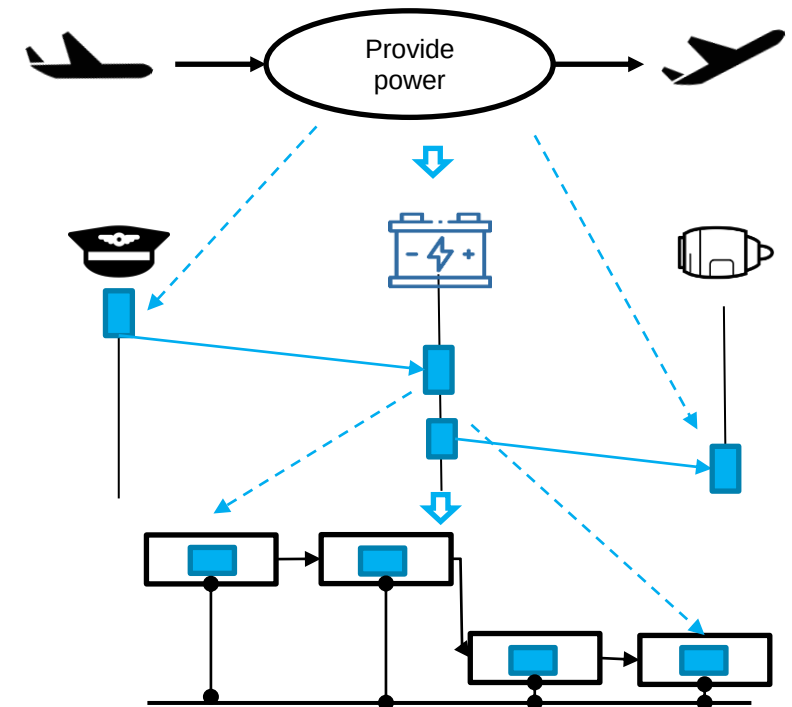
- ✓ Identify client/user needs
- ✓ Describe use-case: activities, interactions, environment
- ✓ Identify and delimit the system of interest

Step 2 – Systems/logical Analysis

- ✓ Identify system functionalities to answer users' needs
- ✓ Define interfaces between the system and external actors
- ✓ Describe system functions, flow of activities, behavior

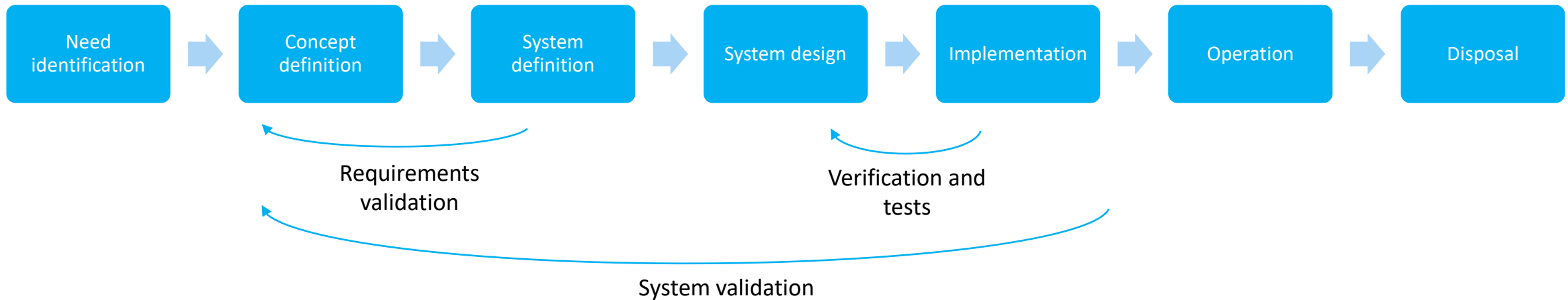
Step 3 – Systems Architecture

- ✓ Design the system to supply the required functionalities
- ✓ Subsystem/component architecture, interfaces, connections...



Fundamentals of systems engineering

System lifecycle – verification and validation



- **Verification** – the system does what it was intended to do
The system is compliant with the requirements
- **Validation** – the system does what the end user wants
The system is compliant with the user needs



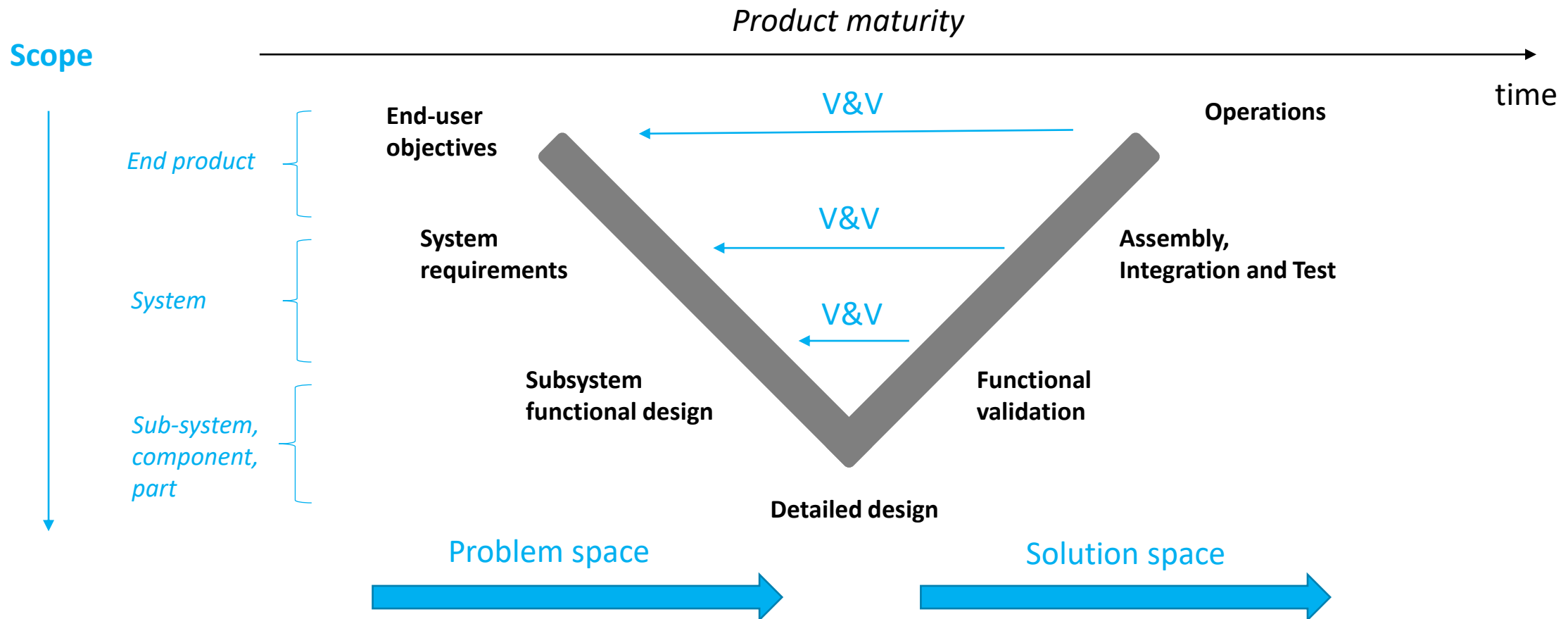
We did what we intended to do



We did what we needed to do

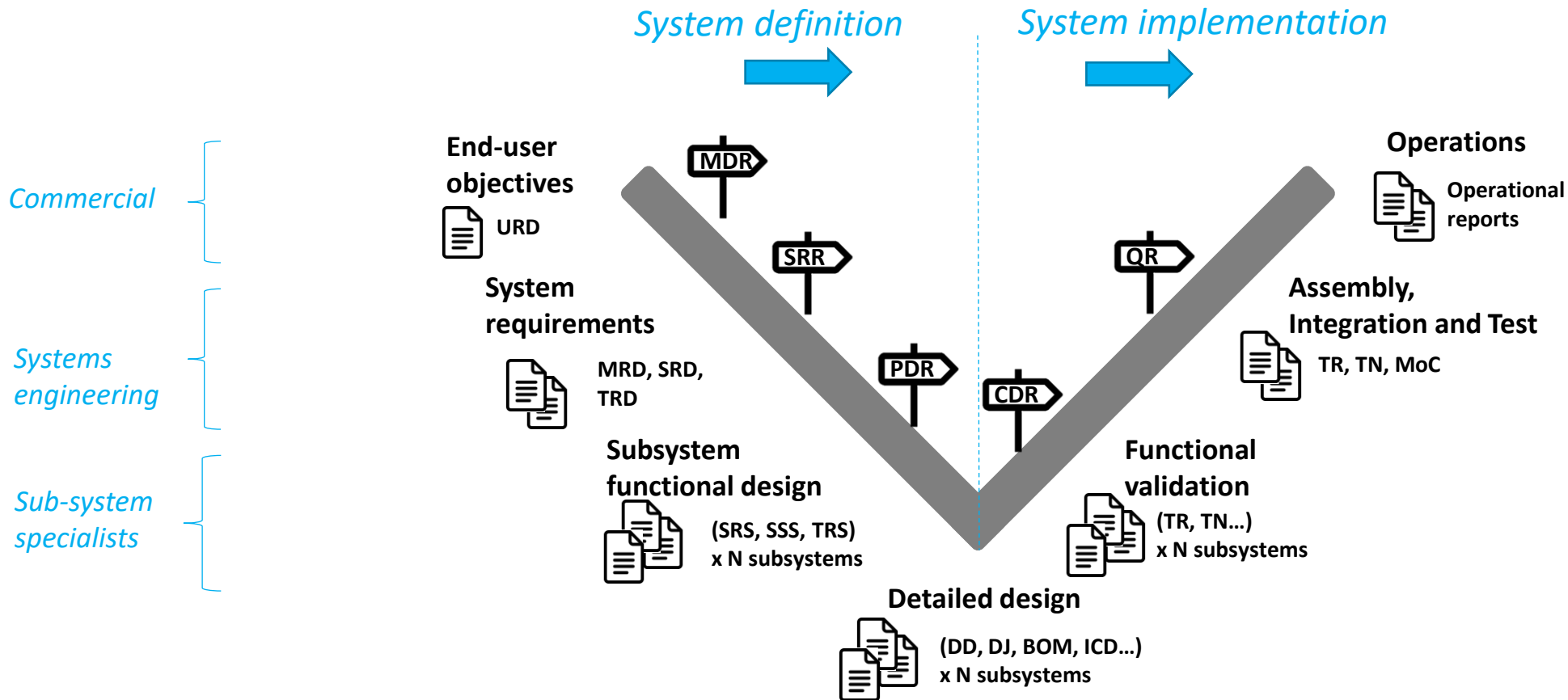
The V-cycle applied to aerospace systems

The *beauty* of the V representation



The V-cycle applied to aerospace systems

Main phases of the development cycle of an aerospace system



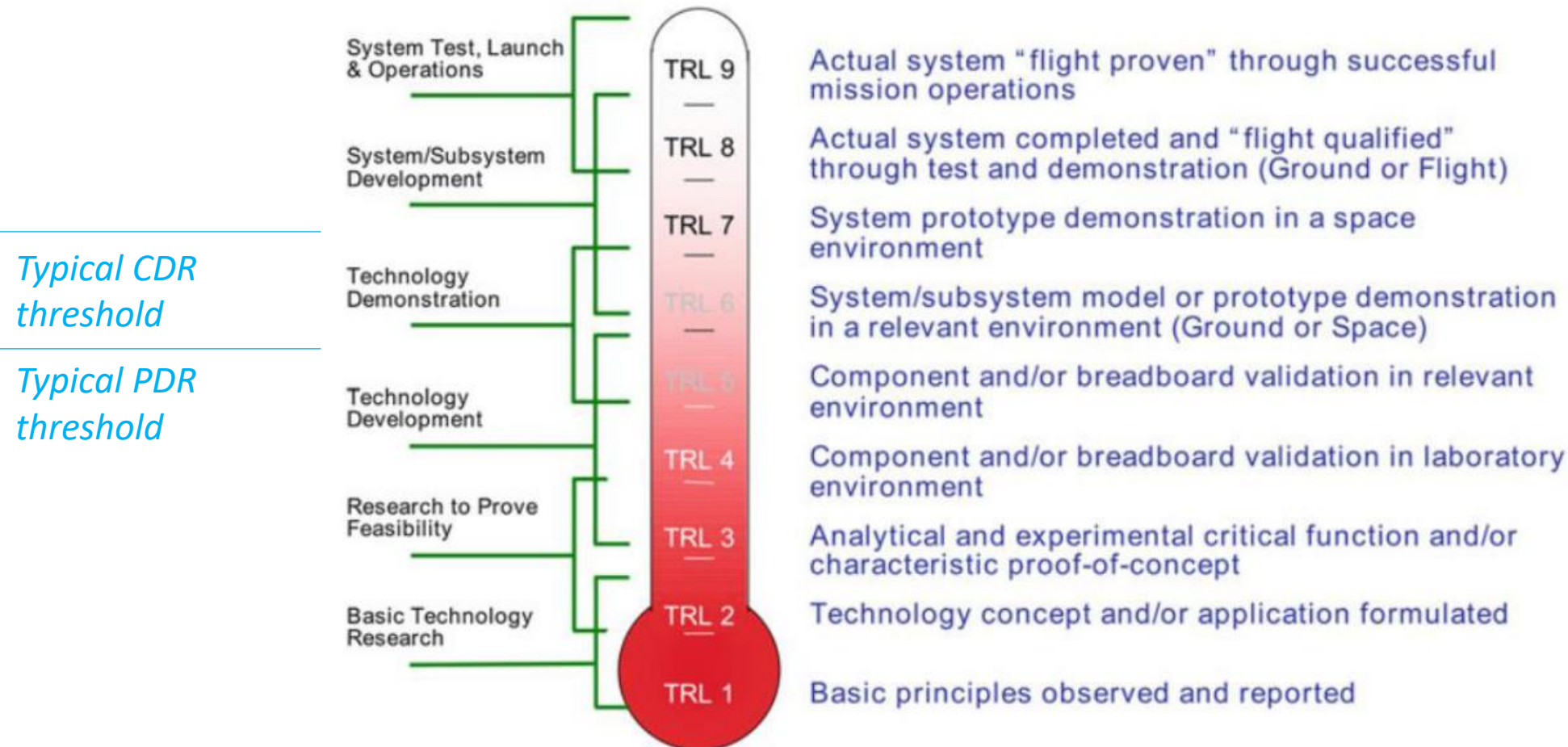
The V-cycle applied to aerospace systems

Main phases of the development cycle of an aerospace system

Phase	Main objective	Closing milestone
Phase 0	Capture end-user needs	MDR – Mission Definition Review
Phase A	Identification of system requirements and conceptual design	SRR – System Requirements Review
Phase B	Preliminary design and identification of technical trade-offs	PDR – Preliminary Design Review
Phase C	Closing technical trade-offs and freeze the solution design	CDR – Critical Design Review
Phase D	Implement the solution: assembly, integration and test	QR – Qualification Review
Phase E	Operations support and maintenance	
Phase F	Disposal	

The V-cycle applied to aerospace systems

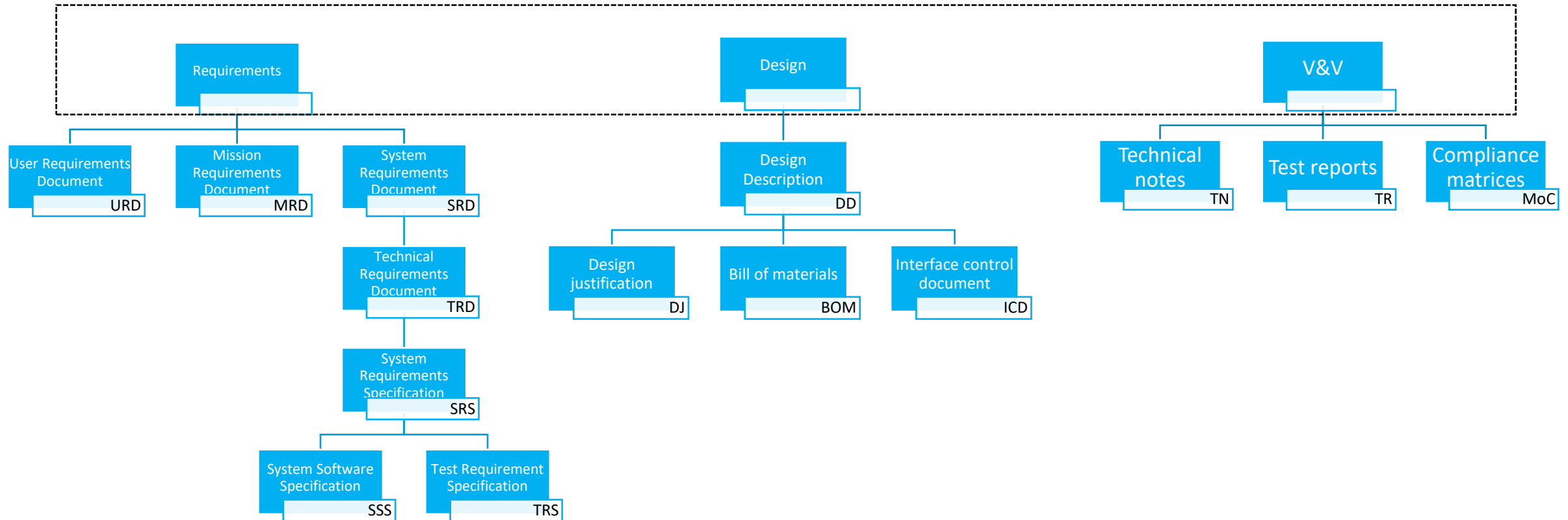
Technology Readiness Level (TRL)



The V-cycle applied to aerospace systems

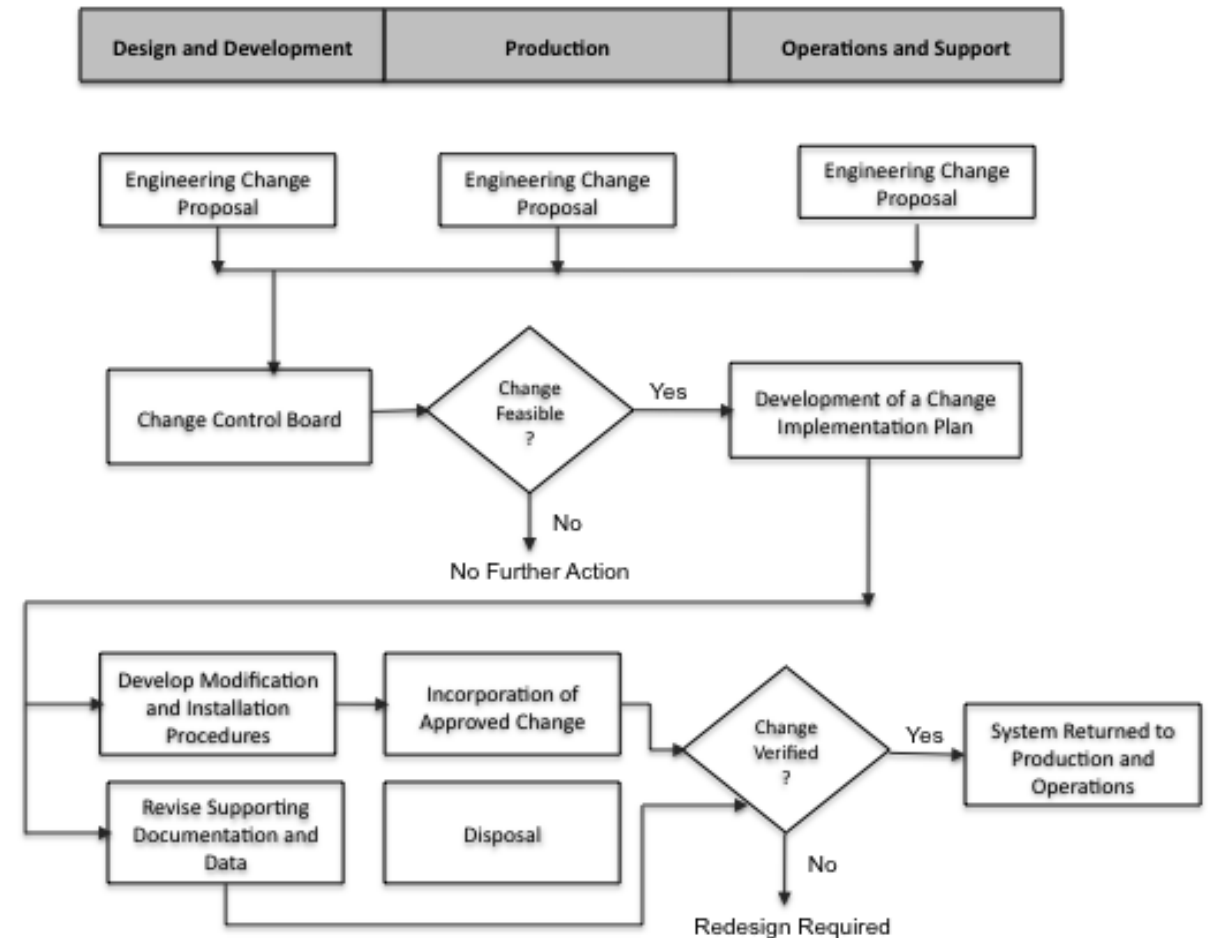
Document tree

For each system / subsystem



Introduction to configuration management

- The purpose of configuration management (CM) is to establish and maintain the integrity of all of the identified outputs of a **project** or **process** and make them available to concerned parties (ISO2015).
- Unmanaged changes to system artifacts (**plans**, **requirements**, **design**, **software**, hardware, testing, and documentation) can lead to problems that persist throughout the system **life cycle**.
- The primary objective of CM is to manage and control the change to such artifacts.



Source: SEBoK - Configuration Management

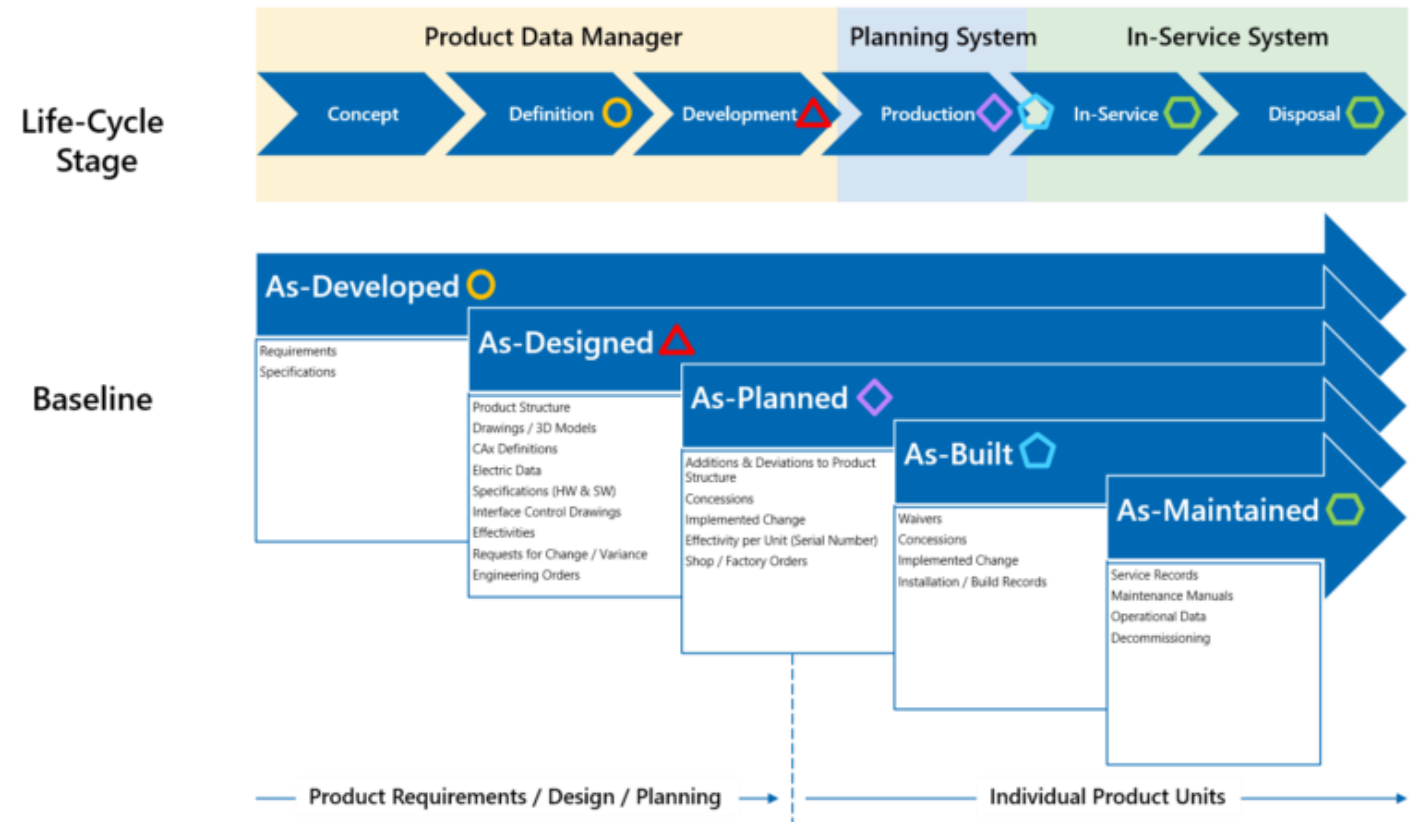
Introduction to configuration management

Baseline

A specification or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures. (ISO/IEEE 2008)

Tools

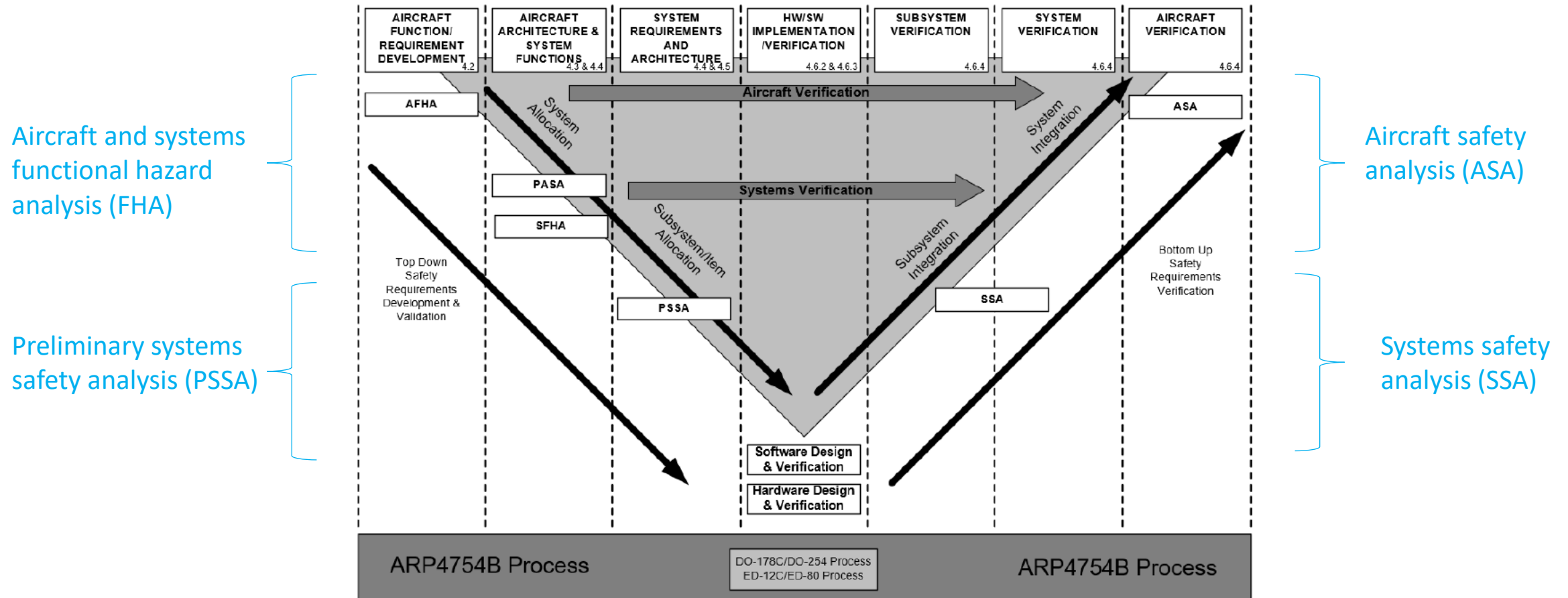
- Library management
- Tracking and change management
- Version management
- Release management



Source: lotar-international

The V-cycle applied to system-safety engineering

Introduction to session 2 – [the safety engineering process](#)

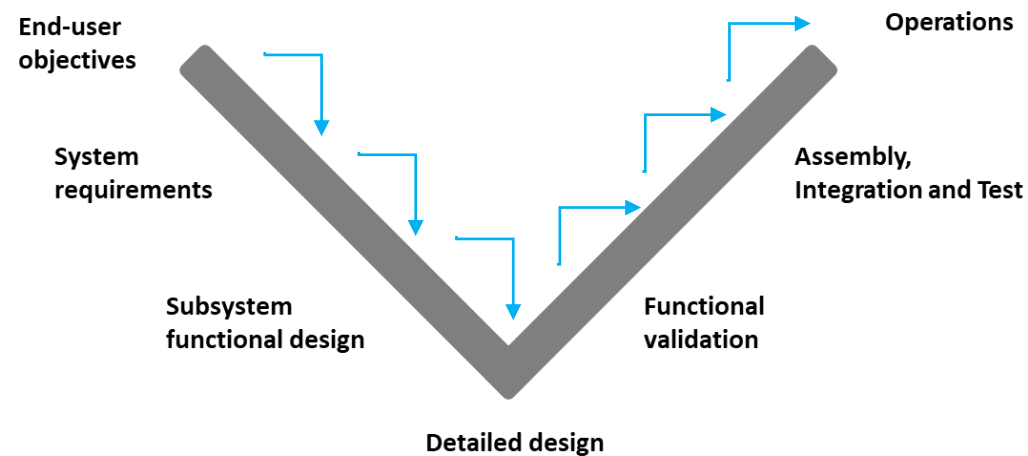


Modern organizations: agile methodology

Waterfall methodology

Development paradigm for complex **physical systems**

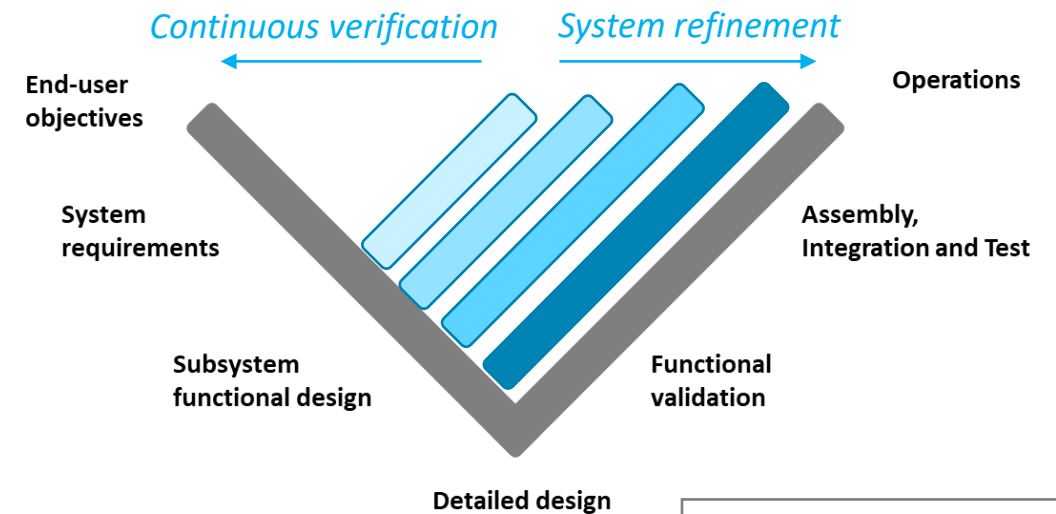
- Sequential steps with increasing level of maturity
- Complete review of the system at each **milestone**
- Based on documentation and expert reviews



Agile methodology

Development paradigm evolving from **software systems**

- System always *ready for service* (with limited features)
- Incremental evolution of the system at each **iteration**
- Based on continuous testing (automated) and user feedback



Key enabler
modeling and simulation

Introduction to systems engineering

Key takeaways

- **Systems engineering** is a key discipline to **master complexity**
- In our context *system* means ***system of interest*** – the system under study
 - The constituents of a system are also systems (with a different scope)
- Systems engineers participate along the whole **lifecycle** of a product
 - From understanding of user needs and conceptual design to the end-product
- The **V-cycle** diagram allows to visualize system decomposition and phases
 - We have presented the main phases of the product lifecycle, and its corresponding design reviews

Any questions?



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